

1

Threat model

- Safety alignment varies across languages; rare, compressed and indirect forms can evade familiar patterns.
- Classical Chinese adds polysemy, allusion, symbolism and nested syntax that conceal sensitive intent.
- The work studies automated black-box red teaming of closed models, not white-box gradients.

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Evaluation

TRANSLATE, FILTER, JUDGE

- A two-stage module translates Classical Chinese responses into English before scoring.
- Keyword detection combines with a GPT-4o semantic-intent judge and manual checks.
- Metrics include attack success rate, toxicity and average victim-model queries.

2

Strategy space

EIGHT PROMPT DIMENSIONS

- Each candidate combines role identity, behavioral guidance, mechanism and metaphor mapping.
- Expression style, knowledge relation, context and trigger pattern complete the Cartesian product.
- A generator maps the harmful query and one strategy point into a Classical Chinese prompt.

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Evidence

HIGH ASR, FEW QUERIES

- On 50 AdvBench prompts, CC-BOS reports 100% ASR across six target models.
- On CLAS and StrongREJECT-small it reports about 98.3-100% ASR, with 1.12-2.38 average queries.
- Claude-3.7 ablation: 18% Chinese alone, 60% with strategy space, 100% with search.

3

CC-BOS search

FRUIT-FLY OPTIMIZATION

- Smell search makes decaying local perturbations; visual search moves toward the best candidate.
- Heavy-tailed Cauchy mutation escapes local optima after stagnation.
- Hash deduplication avoids repeats; population 5 and at most 5 iterations keep query budgets small.

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Security meaning

DEFENSE MUST GENERALIZE

- Input/output Llama-Guard filtering reduces ASR sharply but CC-BOS remains strongest tested.
- Results highlight the need for semantic, multilingual and adaptive defenses.
- Metrics partly rely on an automated judge; closed models change, and the method creates misuse risk.

CC-BOS

Obscure but Effective

Classical Chinese Jailbreak Optimization via Bio-Inspired Search | ICLR 2026

Classical Chinese compresses and obscures malicious intent. CC-BOS systematizes this attack surface into eight prompt dimensions and searches it automatically with a fruit-fly-inspired black-box optimizer.

jailbreak

Classical Chinese

red teaming